DEEPAK KATUKURI

PRODUCT ENGINEER · FULL-STACK · DESIGN + APPLIED AI



CONTACT

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TECHNICAL SKILLS

Languages: Python, JavaScript / TypeScript, SQL, PL/SQL, Java, C++

Frontend & Design: React, Vite, Tailwind CSS, React Query, Recharts, component / design systems, data-dense dashboards, empty / loading / error states, responsive & dark-mode layouts

Backend & APIs: FastAPI, REST APIs (auth, status codes), PostgreSQL, schema design & migrations, background jobs, Docker, Kubernetes

LLM / Applied AI: RAG, retrieval, embeddings, vector search, document processing, agent tools & patterns, prompt engineering, CLIP, DistilBERT, Ollama / OpenAI- & Anthropic-compatible APIs

DevOps & Cloud: Git / GitHub, PR workflows, CI/CD, AWS, object storage, pytest

ML / Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, CNNs, fine-tuning, NumPy, Pandas

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present

BTech – CS & Engineering
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

PL/SQL – Database Management

Core Java – Software Development

OneShield – P&C Insurance

LANGUAGES

English – Fluent

German – Beginner (A1/A2)

PROFILE

Product-minded full-stack engineer and Data Science MSc candidate who owns features from first sketch to production — designing the flow and interface, then building it across the frontend, the backend, and the data behind it. Built a full-stack health product with a React / Vite + Tailwind frontend (data-dense analytics dashboard, considered empty / loading / error states) on a FastAPI + PostgreSQL backend, plus retrieval and document-processing LLM systems from scratch. Comfortable in Python and JavaScript / TypeScript, backed by 2.5 years of production software engineering. Seeking a working-student role to build AI products end-to-end, remote or Munich / Ingolstadt.

PROJECTS

RA Nutrition & Lifestyle App — Full-Stack Product (Design · Frontend · Backend)

React · Vite · Tailwind CSS · React Query · Recharts · FastAPI · PostgreSQL

- Designed and built a health product end-to-end — user flows, React / Vite + Tailwind interface, and a FastAPI + PostgreSQL backend — with JWT authentication + token refresh, meal / symptom logging, and a transparent rule-based recommendation engine.
- Designed a data-dense weekly analytics dashboard (React Query + Recharts) that makes meal and symptom trends readable at a glance, with explicit empty, loading, and error states and a reusable Tailwind component system.
- Modeled normalized data across 4 logical PostgreSQL schemas (core, static, tracking, intelligence); architected a V3 ML tier (scikit-learn / XGBoost + PyTorch LSTM / GRU / TCN) for flare-risk prediction, gated to ship only where it beats the rule-based baseline.

RAG Application — Retrieval & Document Processing from Scratch

Python · Ollama / LLM APIs · sentence-transformers · github.com/Deepak589/RAG_pipeline

- Built a framework-free RAG application in ~270 lines of Python (no LangChain or vector DB) — document chunking, embedding, similarity search, and prompt assembly — integrating local LLM inference via Ollama on the same request / response pattern as OpenAI / Anthropic chat APIs, degrading gracefully when offline.
- Cut repeat embedding cost with a SHA-256-fingerprinted vector cache that auto-invalidates on chunk / model change, pairing a dense bi-encoder retriever (all-MiniLM-L6-v2, cosine top-3) with a TF-IDF baseline to raise relevance; extending to hybrid RRF fusion + cross-encoder reranking.

Food Image-to-Recipe Retrieval — Multimodal AI

PyTorch · CLIP ViT-B/32 · DistilBERT · InfoNCE · MSc final project

- Built a config-driven multimodal retrieval app end-to-end — a dual-encoder (frozen CLIP ViT-B/32 + DistilBERT) mapped into a shared embedding space via symmetric InfoNCE — reaching validation median rank 10/200 on 13.5K images and cutting training ~10x by caching CLIP features, with 58 unit tests. (0.98 ROC-AUC EfficientNetB0 CT-scan classifier also built — PyTorch / timm.)

EXPERIENCE

Software Engineer **ValueMomentum** *Mar 2022 – Sep 2024*

- Owned features across 2 enterprise insurance applications from concept to deployment on OneShield PAS, resolving production defects across policy workflows.
- Built and optimized PL/SQL stored procedures and functions powering backend data operations and endorsement modules serving live policy processing.
- Accelerated releases by running QA / UAT test cases and supporting CI/CD deployment of fixes across the full policy lifecycle.

Machine Learning Intern **SmartBridge × IBM** *Jun – Jul 2019*

- Built and deployed a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) via IBM Watson ML, communicating results with Matplotlib / Seaborn visualizations.